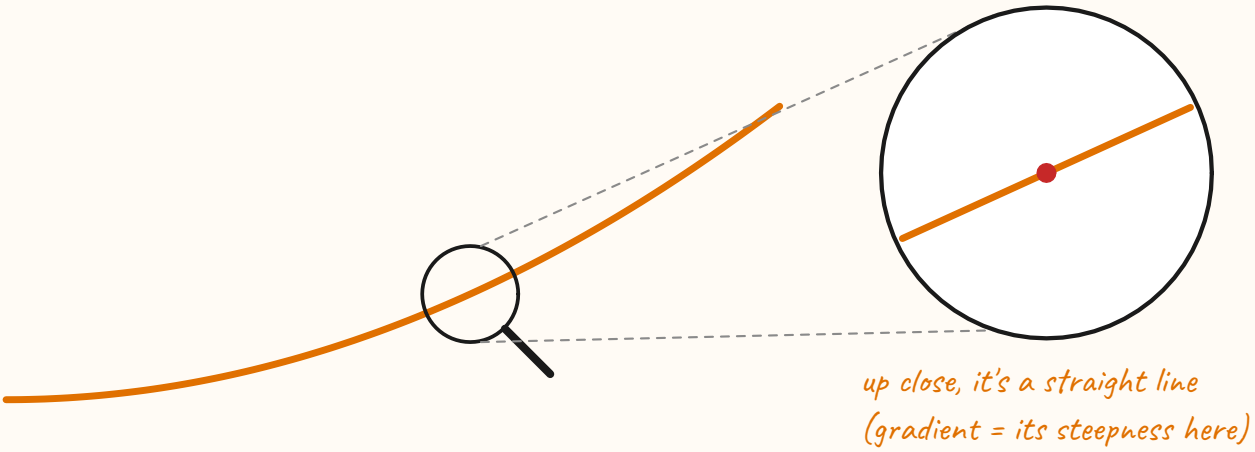
Chapter 34

Gradients of curves

zoom in until it's straight



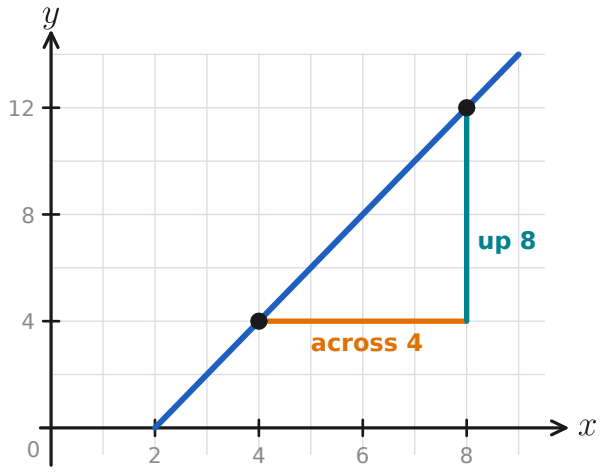
A straight line has the same steepness everywhere (Chapter 20). A curve doesn't: a road over a hill gets steeper, levels off at the top, then tips downwards. This chapter answers **"how steep is it right here?"**

The trick is in the picture: zoom in on any smooth curve and it looks straight, so it has a gradient at that spot. **Differentiation** is the method that finds that gradient for every point at once. It's the first half of calculus.

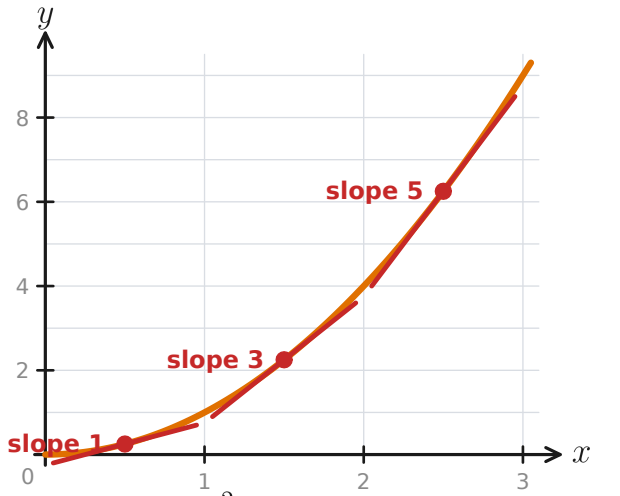
Goes with **Lecture 14** (straight-line recap, tangents, approximating the gradient, differentiation and notation, standard derivatives, stationary points, the second derivative).

A line has one gradient; a curve has many

Recap (Lecture 14): the line through $(4, 4)$ and $(8, 12)$ goes up 8 while it goes across 4, so $m = \frac{8}{4} = 2$. Then $4 = 2 \times 4 + c$ gives $c = -4$: the line is $y = 2x - 4$.



a line: **one** gradient everywhere

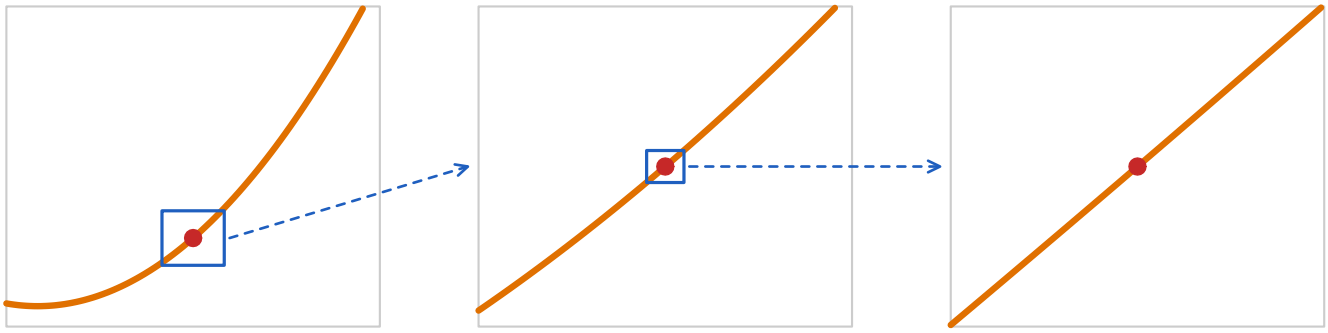


the curve $y = x^2$: a **different** gradient at every point

The gradient of a curve at a point is the gradient of its **tangent** there: the straight line that just touches the curve at that point, heading the same way (the red lines).

Zoom in until it's straight

Lecture 14 zooms in on $y = x^2$. Here's the same idea around the point $(1, 1)$:



×1: clearly curved | **×10: less curved** | **×100: a straight line, gradient 2**

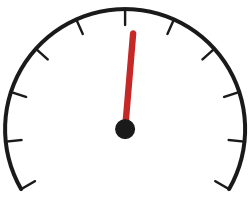
Lecture 14: "The slope that we see is 'roughly' correct, but only for that location!" Zoom in somewhere else and you'd see a line with a different slope.

Average speed vs the speedometer

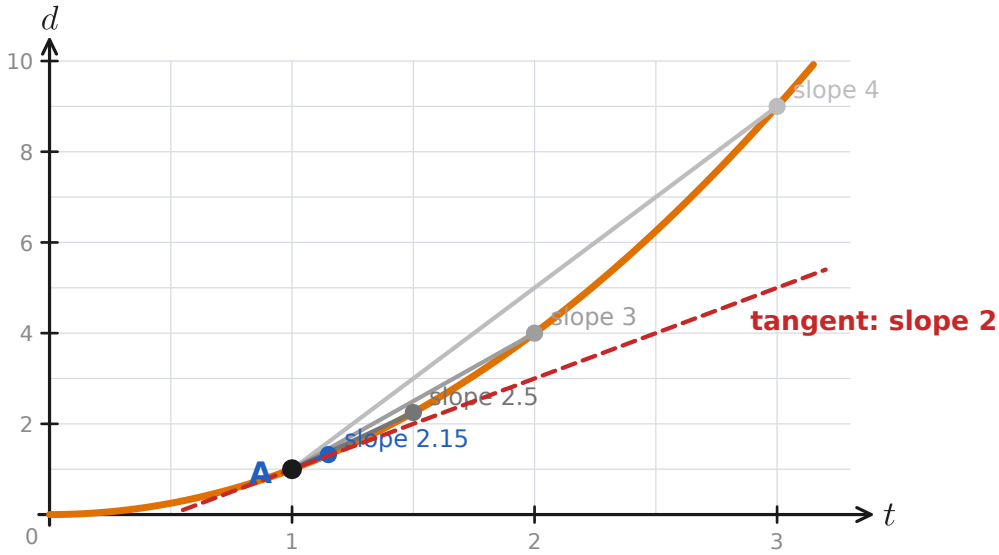
A trip computer says you averaged 40 mph (60 miles in 1.5 hours). The speedometer says what you're doing **right now**. On a distance–time graph:

• the **average** is the gradient of a **chord**: a straight line joining two points of the curve;
• **right now** is the gradient of the **tangent**.

To turn one into the other, slide the second point closer and closer to the first:



right now: 52 mph



The chord slopes 4, 3, 2.5, 2.15 close in on 2: the speed right now at A.

Lecture 14's shrinking step

With numbers: take a small step h along the x axis. The chord from $x = a$ to $x = a + h$ has gradient rise ÷ run:

$$f'(a) \approx \frac{f(a+h) - f(a)}{h}$$ ← *rise ÷ run, and the run is h*

Lecture 14: the gradient of $f(x) = 2x^3 - 3x^2$ at the point $(2, 4)$. (Check the point: $f(2) = 16 - 12 = 4$ ✓.) Shrink h and watch the last column:

step h	f(2 + h)	rise = f(2 + h) − 4	rise ÷ h
1	27	23	**23**
0.1	5.292	1.292	**12.92**
0.01	4.120902	0.120902	**12.0902**
0.001	4.012009	0.012009	**12.009**
0.00001	4.00012	0.000120001	**12.0001**



→ 12

The answers home in on **12**. Now imagine h shrinking **all the way** to nothing. That's **differentiation**, and Lecture 14 lists its two advantages over the table:

1. it gives the gradient at **any** point, not just at $x = 2$;
2. it's **exact** (12, not 12.0001).

The phrasebook: how to read the symbols

$\frac{\mathrm{d}y}{\mathrm{d}x}$ — *"dee y by dee x": the gradient of y as x changes (the derivative)*

$f'(x)$ — *"f dash of x" (or "f prime of x"): the same thing for a function f*

$\delta x,\ \delta y$ — *"delta x": a small step in x (δ is small Greek delta). δy is the matching small step in y*

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ — *"dee two y by dee x squared": differentiate twice (later in this chapter)*

$\frac{\partial y}{\partial x}$ — *"partial y by partial x": when y depends on more than one letter (Chapter 38)*

$\dot{f}(x),\ \dot{y}$ — *"f dot": Newton's dot notation. Lecture 14 writes f(x) for the derivative; most books keep the dot for a rate over time t: ẏ = dy/dt*

Careful: $\frac{\mathrm{d}y}{\mathrm{d}x}$ is **one symbol**, not a fraction of $d \times y$ over $d \times x$. You can't cancel the d's.

The power rule

RULE: the power rule (and constants)

$\frac{\mathrm{d}}{\mathrm{d}x}(ax^n) = anx^{n-1}$: the power comes down and multiplies the number in front, then the power drops by 1. It works for **any** number n: whole, negative or a fraction.

A constant on its own differentiates to **0**: $\frac{\mathrm{d}}{\mathrm{d}x}(7) = 0$.

A term kx differentiates to k: $\frac{\mathrm{d}}{\mathrm{d}x}(5x) = 5$.

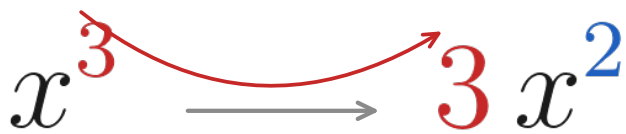
Several terms added or subtracted: differentiate each term on its own and keep the + and − signs.

Every term must look like ax^n **before** you use the rule: rewrite roots and fractions first (box later in this section).

Where the rule comes from. Doing the shrinking-step table every time would be exhausting. So let's measure some gradients (with a tiny h, exactly as above) and hunt for a pattern:

curve	at x = 1	at x = 2	at x = 3	pattern?
$y = x^2$	2	4	6	$2x$
$y = x^3$	3	12	27	$3x^2$
$y = x^4$	4	32	108	$4x^3$

Look at x^3: the gradients 3, 12, 27 are 3×1^2, 3×2^2, 3×3^2. The power **dives to the front**, and what's left has power **one smaller**:



the power jumps down in front...

...and the power goes down by one: 3 − 1 = 2

If a number is already in front, the diving power multiplies it: $5x^4 \to 5 \times 4\,x^3 = 20x^3$. In general:

$$y = ax^n \quad \Rightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = a\,n\,x^{n-1}$$

← *a and n are constants*

A constant has gradient 0: $y = 2$ is a flat line. The rule agrees: $y = 2x^0$ gives $0 \times 2x^{-1} = 0$ (Lecture 14).

Several terms? Differentiate each one separately and keep the + and − signs.

$$y = x \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = 1x^0 = 1$$ ← x is x^1

$$y = 2x + 3 \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = 2 + 0 = 2$$ ← the +3 just lifts the line

$$f(x) = x^3 + 3x \;\Rightarrow\; f'(x) = 3x^2 + 3$$

$$f(z) = 3z^3 - 2z^2 + z \;\Rightarrow\; f'(z) = 9z^2 - 4z + 1$$

any letter works

RULE: rewrite as powers of x before differentiating (Chapter 9)

$\frac{4}{x^3} = 4x^{-3}$: only the x^3 moves up and its power turns negative; the 4 stays a multiplier. It is **not** $(4x)^{-3}$.

$\sqrt{x} = x^{\frac{1}{2}}$, $\sqrt[3]{x} = x^{\frac{1}{3}}$, $x\sqrt{x} = x^1 \times x^{\frac{1}{2}} = x^{\frac{3}{2}}$ (add the powers).

$\frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$, $\frac{3}{2x^2} = \frac{3}{2}x^{-2}$ (numbers on the bottom stay as a fraction in front).

After differentiating you may turn negative and fractional powers back into fractions and roots.

$$y = \frac{4}{x^3} = 4x^{-3} \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = 4(-3)x^{-4} = -\frac{12}{x^4}$$

−3 − 1 = −4

$$y = \frac{1}{x^2} = x^{-2} \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = -2x^{-3} = -\frac{2}{x^3}$$

$$y = x\sqrt{x} = x^{\frac{3}{2}} \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3}{2}x^{\frac{1}{2}} = \frac{3}{2}\sqrt{x}$$ ← 3/2 − 1 = 1/2

$$y = \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{2}x^{-\frac{3}{2}}$$ ← $-\frac{1}{2} - 1 = -\frac{3}{2}$

$$y = \sqrt[4]{x} = x^{\frac{1}{4}} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{4}\,x^{-\frac{3}{4}} = \frac{1}{4x^{3/4}}$$ ← $\frac{1}{4} - 1 = -\frac{3}{4}$

✗ Common mistakes

✗ $\frac{4}{x^3} = (4x)^{-3}$

only the x^3 moves; the 4 stays in front

✓ $\frac{4}{x^3} = 4x^{-3}$

✗ $\frac{\mathrm{d}}{\mathrm{d}x}(5x) = 0$

5x is a line of gradient 5; only a constant on its own gives 0

✓ $\frac{\mathrm{d}}{\mathrm{d}x}(5x) = 5$

✗ $\sqrt{x} \to \frac{1}{2}\sqrt{x}$

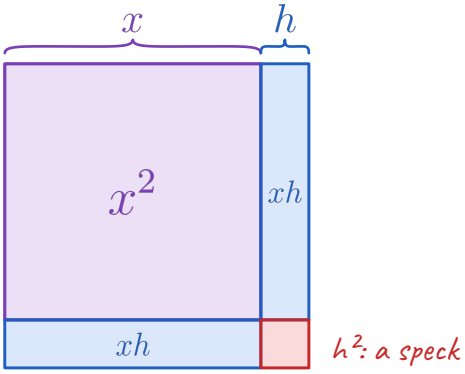
rewrite as a power first, then take 1 off the power

✓ $x^{\frac{1}{2}} \to \frac{1}{2}x^{-\frac{1}{2}}$

Lecture 14's sneaky one: $x = \frac{1}{3}(y+2)$. Make y the subject first (Chapter 14): $y = 3x - 2$, so $\frac{\mathrm{d}y}{\mathrm{d}x} = 3$.

Why the power rule works: grow a square, grow a cube

The table found the pattern; a picture shows **why**. Think of $y = x^2$ as the **area of a square** with side x. Nudge the side up by a tiny amount h. How much new area appears?



(h drawn far bigger than it really is)

New area = two thin strips (x by h each) plus a tiny corner (h by h).

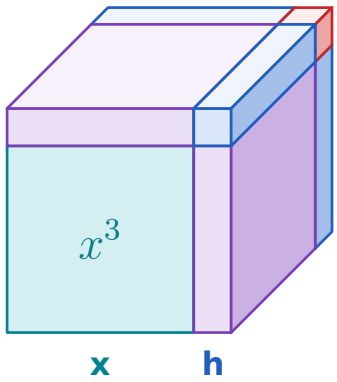
Gradient = new area ÷ nudge:

$$\frac{2xh + h^2}{h} = 2x + h$$

shrink h to nothing and only 2x is left

So the **2 in $2x$ is the two strips**: the square grows along two edges. (It's Chapter 11's $(x+1)^2$ picture with the 1 shrunk down to a tiny h.)

Now $y = x^3$: the volume of a cube. Nudge every edge by h. The cube grows the same way in 3D: three thin **slabs** (x by x by h, one on the top, one on the right, one hidden on the back), three thin **rods** along the edges where slabs meet, and a speck of a cube in the corner:



3 slabs: 3 × x²h

3 rods: 3 × xh²

1 speck: h³

$$\frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2$$

shrink h: the rods and the speck vanish, the slabs stay. Gradient $3x^2$

The **3 in $3x^2$ is the three slabs**: a cube grows on three faces, and each slab has area x^2. You can't draw x^4, but the pattern carries on: n "slabs" of size x^{n-1}, which is exactly the power diving down to the front and dropping by one.

The standard derivatives

Lecture 14's list. Learn these like vocabulary: every harder derivative is built from them. (Angles in radians, Chapter 27.)

function y	derivative dy/dx	
c	0	*flat line*
ax^n	anx^{n-1}	*the power rule*
$\ln x$	$\frac{1}{x}$	*natural log (Chapter 22)*
ae^{kx}	ake^{kx}	*e is its own gradient; the k comes from the inside kx*
$a\sin(bx)$	$ab\cos(bx)$	*the b comes from the inside bx*
$a\cos(bx)$	$-ab\sin(bx)$	*note the minus*
$\tan x$	$\sec^2 x$	*sec x = 1/cos x*

Where the extra b and k come from. In $\sin(bx)$ the angle bx changes b times as fast as x does, so every gradient is b times bigger: $\frac{\mathrm{d}}{\mathrm{d}x}\sin(bx) = b\cos(bx)$. The same happens with $e^{kx} \to ke^{kx}$. This is the **chain rule** (Chapter 35): differentiate the outside function, then multiply by the derivative of the inside ($bx \to b$, $kx \to k$). A number a in front just multiplies the answer.

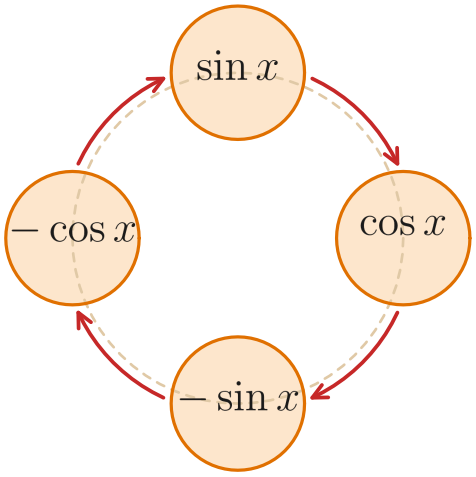
$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(3x) = 3\cos(3x)$$ ← *inside 3x → 3*

$$\frac{\mathrm{d}}{\mathrm{d}x}5e^{-2x} = 5 \times (-2)e^{-2x} = -10e^{-2x}$$ ← *k = −2*

$$\frac{\mathrm{d}}{\mathrm{d}x}4\cos\left(\frac{x}{2}\right) = -4 \times \frac{1}{2}\sin\left(\frac{x}{2}\right) = -2\sin\left(\frac{x}{2}\right)$$

b = ½; cos gives a minus

The sin–cos wheel. Differentiating sine and cosine goes round in a circle. Each step clockwise is one differentiation:



sin → cos → −sin → −cos → back to sin.

Four differentiations bring you home. (Powers vanish instead: $x^3 \to 3x^2 \to 6x \to 6 \to 0$.)

Gradients are rates of change

A gradient is "how fast one thing changes as another changes". With time t instead of x you write $\frac{\mathrm{d}}{\mathrm{d}t}$. Lecture 14: velocity $v = 3t^2 + 2$.

$$\frac{\mathrm{d}v}{\mathrm{d}t} = 6t$$

power rule; the +2 vanishes

$$t = 3: \quad \frac{\mathrm{d}v}{\mathrm{d}t} = 6 \times 3 = 18$$

How fast velocity changes is **acceleration**, so at $t = 3$ the acceleration is 18 (m/s² if v is in m/s). Likewise, the gradient of distance against time is velocity: the speedometer.

The gradient at a point, and finding unknowns

RULE: gradient at a point

The gradient of $y = f(x)$ at $x = a$ is $f'(a)$: differentiate **first**, then put $x = a$ into $\frac{dy}{dx}$.

Never substitute into y first: y at one point is just a number, and a number differentiates to 0.

Given a point (a, b), only the x-value a goes into $\frac{dy}{dx}$.

Example: the gradient of $y = x^3 - 2x^2 + 5$ at the point $(2, 5)$.

$$\frac{dy}{dx} = 3x^2 - 4x$$ ← differentiate first

$$x = 2: \quad 3(2)^2 - 4(2) = 12 - 8 = 4$$ ← then substitute x = 2

PROCEDURE: given the gradient, find the unknown

1. Differentiate, treating any unknown letter (a, k, p...) as an ordinary number.
2. Write "$\frac{dy}{dx}$ at the given x = the given gradient" as an equation.
3. Solve it. If the question asks for a **point**, put each x into the **original** y equation to get its y-coordinate.
4. Check by substituting back.

Example: $y = ax^2 + 3x$ has gradient 7 at $x = 1$. Find a.

$$\frac{dy}{dx} = 2ax + 3$$ ← a is just a number

$$2a(1) + 3 = 7 \Rightarrow 2a = 4 \Rightarrow a = 2$$ ← x = 1, gradient 7

Example: where on $y = x^2 - 4x + 1$ is the gradient 2?

$$2x - 4 = 2 \Rightarrow x = 3$$ ← set dy/dx equal to 2

$$y = 3^2 - 4(3) + 1 = -2$$ ← y from the ORIGINAL equation

The point is **(3, −2)**.

Example: find the points on $y = x^3 - 3x^2$ where the gradient is 9.

$$3x^2 - 6x = 9 \;\Rightarrow\; x^2 - 2x - 3 = 0$$ ← ÷ 3 and rearrange

$$(x-3)(x+1) = 0 \;\Rightarrow\; x = 3 \text{ or } x = -1$$ ← factorise (Chapter 12)

$$y(3) = 27 - 27 = 0, \qquad y(-1) = -1 - 3 = -4$$

original equation

Two points: **(3, 0)** and **(−1, −4)**. A quadratic for x often gives two answers; keep both.

Tangents and normals

RULE: tangent and normal

The **tangent** at a point touches the curve there and has the curve's gradient, $m = f'(a)$.

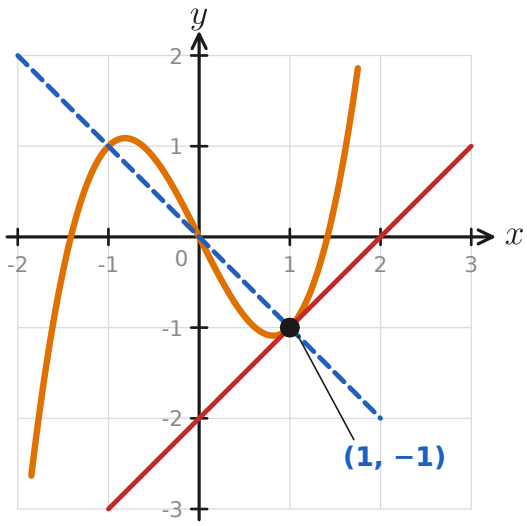
The **normal** is the line through the same point at right angles to the tangent. Its gradient is $-\frac{1}{m}$ (Chapter 20: perpendicular gradients multiply to -1).

Both are straight lines: write them with Chapter 20's point-gradient form $y - y_1 = m(x - x_1)$.

PROCEDURE: equation of the tangent or normal at x = a

1. If only $x = a$ is given, find y_1 by putting a into the **original** equation. The point is (a, y_1).
2. Differentiate.
3. Tangent gradient: $m = f'(a)$ (put a into $\frac{\mathrm{d}y}{\mathrm{d}x}$).
4. Tangent: $y - y_1 = m(x - x_1)$, then tidy into $y = mx + c$ (or $ax + by + c = 0$ if asked).
5. Normal: same point, gradient $-\frac{1}{m}$. (If $m = 0$ the tangent is flat and the normal is the vertical line $x = a$.)

Example: the tangent and normal to $y = x^3 - 2x$ at $x = 1$.



orange: the curve $y = x^3 - 2x$
red: the tangent, gradient 1
blue dashes: the normal, gradient −1

The normal crosses the tangent at a right angle (the grid squares are true squares here).

$$y_1 = 1^3 - 2(1) = -1$$ ← *step 1: the point is (1, −1)*

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 2$$ ← *step 2*

$$m = 3(1)^2 - 2 = 1$$ ← *step 3*

$$y - (-1) = 1(x - 1) \;\Rightarrow\; y = x - 2$$ ← *step 4: tangent*

$$m_N = -\tfrac{1}{1} = -1 : \quad y + 1 = -(x - 1) \;\Rightarrow\; y = -x$$

step 5: normal

Example: the tangent and normal to $y = \frac{2}{x}$ at $(1, 2)$.

$$y = 2x^{-1} \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = -2x^{-2} = -\frac{2}{x^2}$$ ← *rewrite first*

$$m = -\frac{2}{1^2} = -2$$ ← *point given, so skip step 1*

$$y - 2 = -2(x - 1) \;\Rightarrow\; y = -2x + 4$$ ← *tangent*

$$m_N = \tfrac{1}{2} : \quad y - 2 = \tfrac{1}{2}(x - 1) \;\Rightarrow\; y = \tfrac{1}{2}x + \tfrac{3}{2}$$

normal: flip and change sign

As $ax + by + c = 0$ the normal is $x - 2y + 3 = 0$ (multiply by 2 and collect).

✗ Common mistakes

✗ $m = f(a)$ ✓ $m = f'(a)$

the gradient comes from dy/dx, not from y

✗ $m_N = -m$ ✓ $m_N = -\frac{1}{m}$

flip the fraction AND change the sign

✗ y_1 from $\frac{\mathrm{d}y}{\mathrm{d}x}$ ✓ y_1 from $y = f(x)$

the point lies on the curve, so use the curve's equation

Stationary points: hilltops, valleys and ledges

RULE: stationary points

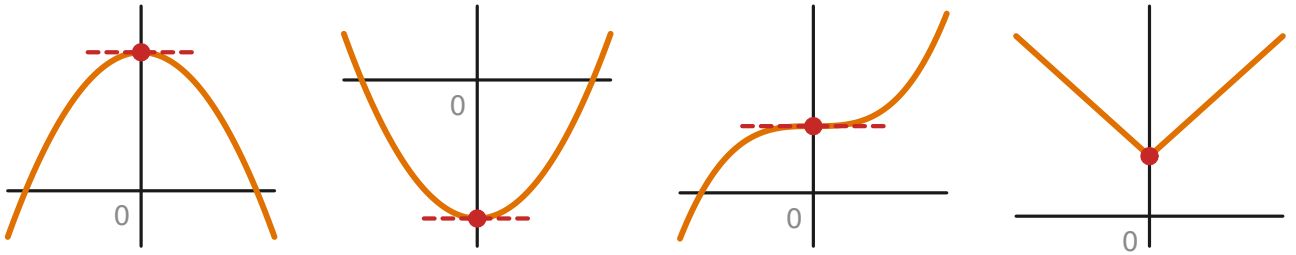
A **stationary point** is where $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$: the tangent is horizontal.

It is a **local maximum** (hilltop), a **local minimum** (valley) or a **stationary point of inflection** (a ledge, where the curve carries on the same way).

PROCEDURE: finding and classifying stationary points

1. Differentiate to get $\frac{\mathrm{d}y}{\mathrm{d}x}$.
2. Set $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$.
3. Solve for x (factorise or use the quadratic formula; there may be more than one answer).
4. Put each x into the **original** y equation to get the y-coordinate. (Not into $\frac{\mathrm{d}y}{\mathrm{d}x}$: that is 0 there by definition.)
5. Classify each one: second-derivative test (next section), or the sign of the gradient just either side: $+, 0, -$ hilltop; $-, 0, +$ valley; same sign both sides: inflection.

Where the gradient is **exactly 0** the curve has stopped rising or falling, for a moment. Lecture 14's gallery:



local maximum (0, 3): stationary and turning

local minimum (0, −3): stationary and turning

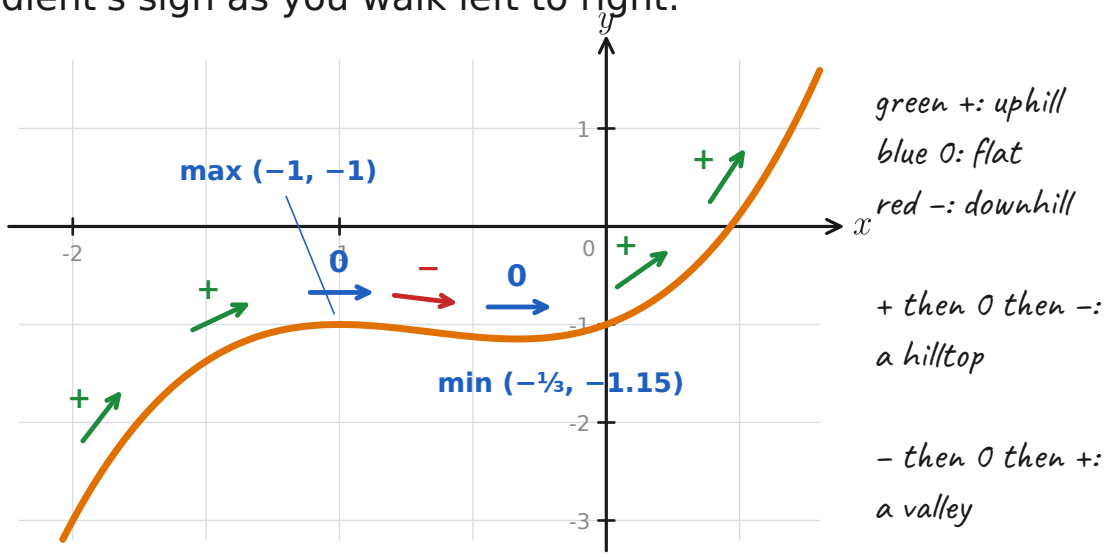
inflection (0, 2): stationary, NOT turning

sharp corner (0, 1): turning, NOT stationary

Lecture 14, by the procedure: $y = -x^2 + 3$. Step 1: $\frac{\mathrm{d}y}{\mathrm{d}x} = -2x$. Steps 2–3: $-2x = 0$ only when $x = 0$. Step 4: $y = -0^2 + 3 = 3$. The stationary point is $(0, 3)$.

Careful with words: the slide says differentiating finds the curve's "zeros". It finds where the **gradient** is zero (stationary points), not where the curve crosses the x axis.

Lecture 14's big example: $y = x^3 + 2x^2 + x - 1$. The arrows show the gradient's sign as you walk left to right:



$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 + 4x + 1 = 0$$ ← set the gradient to zero

$$x = \frac{-4 \pm \sqrt{16 - 12}}{6} = \frac{-4 \pm 2}{6}$$ ← quadratic formula, a = 3, b = 4, c = 1 (Chapter 15)

$$x = -\frac{1}{3} \quad \text{or} \quad x = -1$$

Which kind is $x = -1$? Test the gradient just either side (Lecture 14):

$$x = -1.1 : \; 3(-1.1)^2 + 4(-1.1) + 1 = 3.63 - 4.4 + 1 = 0.23$$

\+ uphill

$$x = -0.9 : \; 3(-0.9)^2 + 4(-0.9) + 1 = 2.43 - 3.6 + 1 = -0.17$$

− downhill

Uphill, flat, downhill: $x = -1$ is a **maximum**, at $y = -1 + 2 - 1 - 1 = -1$.

The second derivative: smile or frown?

RULE: the second-derivative test

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ means: differentiate $\frac{\mathrm{d}y}{\mathrm{d}x}$ again. Also written $f''(x)$.

At a stationary point $x = a$, work out $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ at $x = a$:

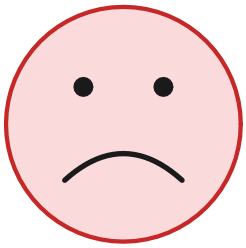
$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} > 0$: **minimum**. $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} < 0$: **maximum**.

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$: the test is **inconclusive** (it could be a maximum, a minimum or an inflection). Use the gradient-sign test either side instead.

Differentiate the derivative and you get the **second derivative** $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$: how fast the **gradient itself** is changing (Lecture 14).



positive: gradient growing
(−, 0, +) → a valley: minimum



negative: gradient shrinking
(+, 0, −) → a hilltop: maximum

Lecture 14 finishes the big example:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 + 4x + 1 \quad \Rightarrow \quad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x + 4 \leftarrow \textit{differentiate again}$$

$$x = -1: \;\; 6(-1) + 4 = -2 < 0 \leftarrow \textit{frown: maximum} \checkmark$$

$$x = -\tfrac{1}{3}: \;\; 6(-\tfrac{1}{3}) + 4 = 2 > 0 \leftarrow \textit{smile: minimum}$$

Why zero decides nothing. $y = x^3$ and $y = x^4$ both have $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ and $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$ at $x = 0$. Yet x^3 has a ledge there (gradient $+, 0, +$) and x^4 has a minimum (gradient $-, 0, +$). Only the either-side test can tell them apart.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Notation

$$\frac{dy}{dx} = f'(x) = \dot{f}(x)$$

the gradient of the tangent; the dot is mostly used for rates over time

Power rule

$$\frac{d}{dx}(ax^n) = anx^{n-1}$$

any n: whole, negative or fraction

Constants, kx, sums

$$\frac{d}{dx}(c) = 0, \qquad \frac{d}{dx}(kx) = k$$

differentiate a sum term by term, keeping the signs

Rewrite first

$$\frac{a}{x^n} = ax^{-n}, \quad \sqrt{x} = x^{\frac{1}{2}}, \quad \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$$

$\frac{4}{x^3} = 4x^{-3}$, not $(4x)^{-3}$

ln and e

$$\frac{d}{dx}\ln x = \frac{1}{x}, \qquad \frac{d}{dx}e^{kx} = ke^{kx}$$

Trig (radians)

$$\frac{d}{dx}\sin bx = b\cos bx, \quad \frac{d}{dx}\cos bx = -b\sin bx, \quad \frac{d}{dx}\tan x = \sec^2 x$$

Gradient at x = a

$$m = f'(a)$$

differentiate first, then substitute. Unknown constant or x: set $f'(a)$ equal to the given gradient and solve

Tangent at x = a

$$y - y_1 = f'(a)(x - x_1)$$

point $(a, f(a))$

Normal at x = a

$$m_N = -\frac{1}{f'(a)}$$

same point, perpendicular gradient

Stationary points

$$\frac{dy}{dx} = 0$$

solve for x; get y from the original equation

Second-derivative test

$$\frac{d^2y}{dx^2} > 0 \text{ min}, \quad < 0 \text{ max}, \quad = 0 \text{ test fails}$$

if it fails, check the gradient's sign either side

Your turn

Answers in grey. For stationary points, sketch the curve afterwards: does your answer match the picture?

Gradient from scratch

1. Estimate the gradient of $y = x^2$ at $x = 3$ using $h = 0.01$.

1. $(9.0601 - 9) \div 0.01 = 6.01$ (the exact answer is 6)

Power rule

2. $y = 5x^4$
3. $y = 3x^2 - 7x + 2$
4. $y = \sqrt{x}$
5. $y = 4/x^3$

2. $20x^3$ 3. $6x - 7$ 4. $\frac{1}{2}x^{(-\frac{1}{2})} = 1/(2\sqrt{x})$ 5. $-12x^{-4} = -12/x^4$

Standard derivatives

6. $y = 3e^{(2x)}$
7. $y = 2 \sin 5x$
8. the gradient of $y = \ln x$ at $x = 4$

6. $6e^{(2x)}$ 7. $10 \cos 5x$ 8. $1/4 = 0.25$

Stationary points

9. $y = x^2 - 6x + 1$: find the stationary point and say what kind it is.
10. $y = x^3 - 3x$: find both stationary points and classify them with the second derivative.

9. $x = 3$, $y = -8$: minimum (second derivative $2 > 0$) 10. $(1, -2)$ minimum ($6 > 0$); $(-1, 2)$ maximum ($-6 < 0$)

Tangents and normals

11. Find the tangent to $y = x^2 + 2x$ at $x = 1$.
12. Find the normal to $y = x^2 + 2x$ at $x = 1$.
13. Find the tangent to $y = 6/x$ at $(2, 3)$.

11. point $(1, 3)$, $m = 4$: $y = 4x - 1$ 12. $m = -1/4$: $y = -x/4 + 13/4$ 13. $dy/dx = -6/x^2 = -3/2$: $y = -3x/2 + 6$

Find the unknown

14. $y = ax^2 + 3x$ has gradient 7 at $x = 1$. Find a.

15. Find the point on $y = x^2 - 5x + 2$ where the gradient is 3.

16. $y = x^3 + kx$ has a stationary point at $x = 2$. Find k.

17. Find the points on $y = x^3 - 12x + 1$ where the gradient is 15.

18. The tangent to $y = x^2 + bx$ at $x = 2$ has gradient 1. Find b, then the tangent.

14. $a = 2$ ($2a + 3 = 7$) 15. $(4, -2)$ ($2x - 5 = 3$, so $x = 4$) 16. $k = -12$ ($3(2)^2 + k = 0$) 17. $(3, -8)$ and $(-3, 10)$ ($3x^2 - 12 = 15$, so $x^2 = 9$) 18. $b = -3$ ($4 + b = 1$); point $(2, -2)$, tangent $y = x - 4$